

EXPERIMENT NO. 9

Compute Cohesion and Coupling for the Project

Project: Hospital Management System (HMS)

1. Aim

To identify the cohesion type of each major module and to compute the coupling value between selected module pairs of the Hospital Management System (HMS).

2. Objectives

- To analyze software design quality using the factors of Cohesion and Coupling.
- To classify each HMS module by its cohesion type and justify the classification.
- To compute a quantitative coupling value for selected module interactions.

3. Software Required

Any text editor / open source tool.

4. Cohesion Analysis

Cohesion refers to how closely the responsibilities within a single module are related. High cohesion (Functional, Sequential) is desirable; lower cohesion (Logical, Temporal, Coincidental) indicates a module that could be refactored.

Module	Description	Cohesion Type	Justification
Patient Registration Module	Performs a single well-defined task: capturing patient details and generating a Patient ID.	Functional	High cohesion - all statements contribute to one function.
Appointment Scheduling Module	Dedicated solely to checking availability and booking/confirming appointments.	Functional	High cohesion - single clear purpose.
Billing Module	Calculates the bill and the output (bill amount) is immediately used as input to process payment.	Sequential	Output of one activity feeds the next within the module.
Pharmacy Module	Check-stock, dispense-medicine, and update-stock operations all act on the same medicine stock data.	Communicational	All functions operate on the same data structure (stock).
Notification Module	Groups SMS, email, and in-app notice functions selected by a type flag; functions are related only by category, not by data or sequence.	Logical	Lower cohesion - functions grouped by category, chosen via a control flag.
Report Generation Module	Generates patient-inflow, revenue, and occupancy reports, typically triggered together at the end of a period.	Temporal	Lower cohesion - functions related mainly by when they execute.

5. Coupling Computation

Coupling refers to the degree of interdependence between modules. The Module Coupling metric used is:

$$C = 1 - 1 / (1 + di + 2ci + do + 2co + gd + 2gc + w + r)$$

where di = input data parameters, ci = input control parameters, do = output data parameters, co = output control parameters, gd = global variables used as data, gc = global variables used as control, w = fan-out (modules called), r = fan-in (modules calling it). A value of C closer to 1.0 indicates higher coupling; a lower value indicates looser coupling.

Module Pair	di	ci	do	co	gd	gc	w	r	Sum	C
Patient Registration -> PatientDB	2	0	1	0	0	0	1	1	5	0.833
AppointmentController -> BillingService	2	1	1	0	0	0	1	2	8	0.889
Pharmacy (Dispense) -> MedicineStock	1	1	1	1	2	1	1	1	12	0.923
NotificationModule -> Recipients	1	0	0	0	0	0	1	3	5	0.833

Observation: The Pharmacy (Dispense) -> MedicineStock pair shows the highest coupling value (0.923) because it relies on shared global stock data (gd, gc) in addition to explicit parameters. The Patient Registration and Notification modules show comparatively lower coupling (0.833) as they interact mainly through explicit data parameters with minimal global/control dependence.

6. Conclusion

Most HMS modules exhibit Functional or Sequential cohesion, which is desirable, while the Notification and Report Generation modules show Logical/Temporal cohesion and are candidates for refactoring into more focused sub-modules. The coupling analysis shows that the Pharmacy module's dependence on shared global stock data makes it the most tightly coupled; replacing global stock access with explicit method parameters would reduce coupling and improve maintainability, consistent with the general software design goal of high cohesion and low coupling.